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Studierichting: Informatica
Oefeningenreeks 4A nr. 15.11

$$\int x^2 \cdot \cosh x \cdot dx$$

$$\begin{cases} u = x^2 \\ dv = \cosh x \cdot dx \end{cases} \Rightarrow \begin{cases} du = 2x \cdot dx \\ v = \sinh x \end{cases}$$

$$= x^2 \sinh x - \int 2x \sinh x \cdot dx$$

$$\begin{cases} u = 2x \\ dv = \sinh x \cdot dx \end{cases} \Rightarrow \begin{cases} du = 2 \cdot dx \\ v = \cosh x \end{cases}$$

$$= x^2 \sinh x - (2x \cosh x - \int 2 \cosh x \cdot dx)$$

$$= x^2 \sinh x - 2x \cosh x + 2 \int \cosh x \cdot dx$$

$$= x^2 \sinh x - 2x \cosh x + 2 \sinh x + C$$

$$= (x^2 + 2) \sinh x - 2x \cosh x + C$$

References